Algebra Problems from Previous Years' Undergraduate Mathematics Olympiads

Selected Important Questions

Polynomials and Equations

1. (2018) Let $f(x) = \sin x - \cos x$. State whether f is increasing/decreasing, concave up/down in the interval $[-,]$. Sketch the graph.

   Solution: First derivative: $f'(x) = \cos x + \sin x = \sqrt{2}\sin(x + /4)$

   Critical points when $\sin(x + /4) = 0$: $x = -\pi/4, 3\pi/4$ in $[-,]$

   - Increasing on $(-/4, 3/4)$ - Decreasing on $[-, -/4) \cup (3/4,]$

   Second derivative: $f''(x) = -\sin x + \cos x = \sqrt{2}\cos(x + /4)$

   Inflection points when $\cos(x + /4) = 0$: $x = -3/4, /4$

   - Concave up on $(-3/4, /4)$ - Concave down on $[-, -3/4) \cup (/4,]$

2. (2018) Use determinants to find the equation of a circle that passes through (2,6), (6,4), (7,1). Hence sketch the circle.

   Solution: General circle equation: $x^2 + y^2 + Dx + Ey + F = 0$

   Using the points:

$$
\begin{aligned}
(2,6): &\quad 4 + 36 + 2D + 6E + F = 0 \Rightarrow 2D + 6E + F = -40\\
(6,4): &\quad 36 + 16 + 6D + 4E + F = 0 \Rightarrow 6D + 4E + F = -52\\
(7,1): &\quad 49 + 1 + 7D + E + F = 0 \Rightarrow 7D + E + F = -50
\end{aligned}
$$

   Solve the system: Subtract first from second: $4D - 2E = -12 \Rightarrow 2D - E = -6$ Subtract second from third: $D - 3E = 2$

   Solve: $D = -4, E = -2, F = -20$

   Circle equation: $x^2 + y^2 - 4x - 2y - 20 = 0$ Center: (2,1), Radius: $\sqrt{4 + 1 + 20} = 5$

3. (2018) Solve for x: $|x + 1| - |x| + 2|x - 1| = 2x - 1$

   Solution: Consider intervals:

   Case 1: $x < -1$ $-(x + 1) + x - 2(x - 1) = 2x - 1$ $-x - 1 + x - 2x + 2 = 2x - 1$ $-2x + 1 = 2x - 1 \Rightarrow 4x = 2 \Rightarrow x = 0.5$ (not in interval)

   Case 2: $-1 \leq x < 0$ $(x + 1) + x - 2(x - 1) = 2x - 1$ $x + 1 + x - 2x + 2 = 2x - 1$ $3 = 2x - 1 \Rightarrow 2x = 4 \Rightarrow x = 2$ (not in interval)

   Case 3: $0 \leq x < 1$ $(x + 1) - x - 2(x - 1) = 2x - 1$ $x + 1 - x - 2x + 2 = 2x - 1$ $-2x + 3 = 2x - 1 \Rightarrow 4x = 4 \Rightarrow x = 1$ (not in interval)

Case 4: $x \geq 1$ $(x+1) - x + 2(x-1) = 2x - 1$ $x + 1 - x + 2x - 2 = 2x - 1$ $2x - 1 = 2x - 1$ (identity)

Solution: $x \geq 1$

4. (2018) The roots of $x^2 + qx + r = 0$ are α, β, γ; Find the equation whose roots are $\beta^2 + \beta\gamma + \gamma^2, \gamma^2 + \gamma\alpha + \alpha^2, \alpha^2 + \alpha\beta + \beta^2$.

   Solution: By Vieta: $\alpha + \beta + \gamma = 0$, $\alpha\beta + \beta\gamma + \gamma\alpha = q$, $\alpha\beta\gamma = -r$

   Note: $\beta^2 + \beta\gamma + \gamma^2 = (\beta + \gamma)^2 - \beta\gamma = \alpha^2 - \beta\gamma$

   But $\beta\gamma = \frac{\alpha\beta\gamma}{\alpha} = \frac{-r}{\alpha}$

   So $\beta^2 + \beta\gamma + \gamma^2 = \alpha^2 + \frac{r}{\alpha}$

   Similarly for others. Let $t = \alpha^2 + \frac{r}{\alpha}$

   Then $\alpha t = \alpha^3 + r$. But $\alpha^3 = -q\alpha - r$, so $\alpha t = -q\alpha$

   Thus $t = -q$ (if $\alpha \neq 0$)

   So all new roots equal $-q$. Required equation: $(x + q)^3 = 0$

5. (2018) When polynomial $P(x)$ is divided by $x-1$, remainder is 1, and when divided by $x+1$, remainder is -1. Find remainder when $P(x)$ is divided by $x^2 - 1$.

   Solution: Let $P(x) = (x^2 - 1)Q(x) + ax + b$

   Then: $P(1) = a + b = 1$ $P(-1) = -a + b = -1$

   Solving: $a = 1, b = 0$

   Remainder: x

Inequalities

6. (2018) If $x, y, z > 0$, prove: (i) $\frac{2}{x+y} + \frac{2}{y+z} + \frac{2}{z+x} > \frac{9}{x+y+z}$ (ii) $xyz > (y + z - x)(z + x - y)(x + y - z)$

   Solution: (i) By Cauchy-Schwarz:

$$[(x+y) + (y+z) + (z+x)]\left(\frac{1}{x+y} + \frac{1}{y+z} + \frac{1}{z+x}\right) \geq 9$$

$$2(x+y+z)\left(\frac{1}{x+y} + \frac{1}{y+z} + \frac{1}{z+x}\right) \geq 9$$

   Multiply by 2: $\frac{2}{x+y} + \frac{2}{y+z} + \frac{2}{z+x} \geq \frac{9}{x+y+z}$

   (ii) Let $a = y+z-x, b = z+x-y, c = x+y-z$. By triangle inequality, $a, b, c > 0$

   Then $x = \frac{b+c}{2}, y = \frac{c+a}{2}, z = \frac{a+b}{2}$

   So $xyz = \frac{(b+c)(c+a)(a+b)}{8}$

   By AM-GM: $(b+c)(c+a)(a+b) \geq 8abc$

   Thus $xyz \geq abc = (y+z-x)(z+x-y)(x+y-z)$

7. (2016) Suppose $a, b > 0$ with $a + b = 1$. Prove:

$$\left(1 + \frac{1}{a}\right)^2 + \left(1 + \frac{1}{b}\right)^2 \geq 18$$

When does equality hold?

Solution: By Cauchy-Schwarz:

$$\left[\left(1 + \frac{1}{a}\right)^2 + \left(1 + \frac{1}{b}\right)^2\right](1^2 + 1^2) \geq \left[\left(1 + \frac{1}{a}\right) + \left(1 + \frac{1}{b}\right)\right]^2$$

But $1 + \frac{1}{a} + 1 + \frac{1}{b} = 2 + \frac{1}{a} + \frac{1}{b}$

By AM-HM: $\frac{1}{a} + \frac{1}{b} \geq \frac{4}{a+b} = 4$

So RHS $\geq (2 + 4)^2 = 36$

Thus $2\left[\left(1 + \frac{1}{a}\right)^2 + \left(1 + \frac{1}{b}\right)^2\right] \geq 36$

So $\left(1 + \frac{1}{a}\right)^2 + \left(1 + \frac{1}{b}\right)^2 \geq 18$

Equality when $a = b = \frac{1}{2}$

Sequences and Series

8. (2018) Let (x_n) be nonzero reals such that:

$$x_n^2 - x_{n-1}x_{n+1} = 1 \quad \text{for } n = 1, 2, 3, \ldots$$

Prove there exists real a such that $x_{n+1} = ax_n - x_{n-1}$ for all $n \geq 1$.

Solution: From given: $x_{n+1} = \frac{x_n^2 - 1}{x_{n-1}}$

We want $x_{n+1} = ax_n - x_{n-1}$

So $\frac{x_n^2 - 1}{x_{n-1}} = ax_n - x_{n-1}$

Multiply: $x_n^2 - 1 = ax_nx_{n-1} - x_{n-1}^2$

Rearrange: $x_n^2 + x_{n-1}^2 - 1 = ax_nx_{n-1}$

So $a = \frac{x_n^2 + x_{n-1}^2 - 1}{x_nx_{n-1}}$

Check this is constant by showing it's independent of n.

For $n = 1$: $a = \frac{x_1^2 + x_0^2 - 1}{x_1x_0}$

For $n = 2$: $a = \frac{x_2^2 + x_1^2 - 1}{x_2x_1}$

From given: $x_1^2 - x_0x_2 = 1 \Rightarrow x_2 = \frac{x_1^2 - 1}{x_0}$

Substitute and verify both expressions equal.

9. (2018) Find sum to n terms: $1 + 2 + 3 + 6 + 17 + 54 + 171 + \cdots$

Solution: Observe pattern: Each term 3 times previous term.

Check: $2 = 3 \times 1 - 1$, $3 = 3 \times 2 - 3$, $6 = 3 \times 3 - 3$, $17 = 3 \times 6 - 1$, $54 = 3 \times 17 - (-3)$

Not consistent. Try recurrence: $a_n = 3a_{n-1} - a_{n-2}$

Check: $3 = 3 \times 2 - 1 = 5$ (no)

Given complexity, likely requires generating function approach.

10. (2018) Consider series $\sum_{k=1}^{\infty} \frac{\sqrt{k+1}-\sqrt{k}}{\sqrt{k^2+k}}$ (a) Find closed form for nth partial sum (b) Determine convergence

    Solution: Simplify term:

$$\frac{\sqrt{k+1}-\sqrt{k}}{\sqrt{k(k+1)}} = \frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}}$$

    So partial sum:

$$S_n = \sum_{k=1}^{n} \left(\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}} \right) = 1 - \frac{1}{\sqrt{n+1}}$$

    As $n \to \infty$, $S_n \to 1$, so series converges to 1.

Functional Equations

11. (2016) Let $f : \mathbb{R} \to \mathbb{R}$ satisfy $f(0) = 1$ and:

$$f(xy+1) = f(x)f(y) - f(y) - x + 2 \quad \text{for all } x, y \in \mathbb{R}$$

    Find $f(x)$.

    Solution: Set $y = 0$: $f(1) = f(x)f(0) - f(0) - x + 2$ But $f(0) = 1$, so $f(1) = f(x) - 1 - x + 2 = f(x) - x + 1$

    Thus $f(x) = f(1) + x - 1$

    Let $f(1) = c$, then $f(x) = x + c - 1$

    Now set $x = 1, y = 1$: LHS: $f(2) = 2 + c - 1 = c + 1$ RHS: $f(1)f(1) - f(1) - 1 + 2 = c^2 - c + 1$

    So $c + 1 = c^2 - c + 1 \Rightarrow c^2 - 2c = 0 \Rightarrow c(c-2) = 0$

    If $c = 0$, $f(x) = x - 1$, check in original equation works.

    If $c = 2$, $f(x) = x + 1$, also works.

    So solutions: $f(x) = x - 1$ or $f(x) = x + 1$

12. (2018) Let f be bijective on $(-1, 1)$ defined by:

$$f(x) = \frac{x}{1-x^2}$$

    Find formula for $f^{-1}(x)$ that works for every real x.

    Solution: Let $y = \frac{x}{1-x^2}$. Then $y(1-x^2) = x \Rightarrow y - yx^2 - x = 0$

    So $yx^2 + x - y = 0$

    Solve: $x = \frac{-1 \pm \sqrt{1+4y^2}}{2y}$ for $y \neq 0$

Since $x \in (-1, 1)$, we take appropriate branch.

For $y > 0$, we need $x \in (0, 1)$, so:

$$f^{-1}(x) = \frac{-1 + \sqrt{1 + 4x^2}}{2x} \quad (x \neq 0)$$

and $f^{-1}(0) = 0$

Domain: all real numbers.

Complex Numbers

13. (2017) Prove:
$$|z + z'|^2 + |z - z'|^2 = 2|z|^2 + 2|z'|^2$$
for any complex numbers z, z'.

    Solution: Let $z = a + bi, z' = c + di$

    Then: $|z + z'|^2 = (a + c)^2 + (b + d)^2$ $|z - z'|^2 = (a - c)^2 + (b - d)^2$

    Sum: $(a^2+2ac+c^2+b^2+2bd+d^2)+(a^2-2ac+c^2+b^2-2bd+d^2) = 2a^2+2b^2+2c^2+2d^2 = 2|z|^2 + 2|z'|^2$

14. (2016) Compute $(1 + i)^{100}$

    Solution: $1 + i = \sqrt{2}e^{i\pi/4}$

    So $(1 + i)^{100} = (\sqrt{2})^{100}e^{i100\pi/4} = 2^{50}e^{i25\pi}$

    Since $e^{i25\pi} = \cos 25\pi + i \sin 25\pi = -1$

    Thus $(1 + i)^{100} = -2^{50}$

Matrix Algebra

15. (2018) Find all pairs (a, b) of real numbers for which there exists unique symmetric 2×2 matrix M with real entries satisfying trace$(M) = a$ and $\det(M) = b$.

    Solution: Let $M = \begin{bmatrix} x & y \\ y & z \end{bmatrix}$

    Then trace$(M) = x + z = a$, $\det(M) = xz - y^2 = b$

    So $z = a - x$, and $x(a - x) - y^2 = b \Rightarrow x^2 - ax + (y^2 + b) = 0$

    For unique M, this quadratic in x must have unique solution for each y, so discriminant zero:
$$\Delta = a^2 - 4(y^2 + b) = 0 \Rightarrow y^2 = \frac{a^2}{4} - b$$

    For unique y, we need $\frac{a^2}{4} - b = 0 \Rightarrow a^2 = 4b$

    Then $y = 0$, $x = a/2$, $z = a/2$

    So unique matrix: $M = \frac{a}{2}I$

16. (2016) Find characteristic polynomial of:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{pmatrix}$$

and hence find its inverse.

Solution: Characteristic polynomial:

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 2 & 3 \\ 2 & 5-\lambda & 3 \\ 1 & 0 & 8-\lambda \end{vmatrix}$$

Expand along row 3:

$$= 1 \cdot \begin{vmatrix} 2 & 3 \\ 5-\lambda & 3 \end{vmatrix} - 0 + (8-\lambda) \begin{vmatrix} 1-\lambda & 2 \\ 2 & 5-\lambda \end{vmatrix}$$

$$= (6 - 3(5-\lambda)) + (8-\lambda)[(1-\lambda)(5-\lambda) - 4]$$
$$= (3\lambda - 9) + (8-\lambda)(\lambda^2 - 6\lambda + 1)$$
$$= -\lambda^3 + 14\lambda^2 - 46\lambda - 1$$

So characteristic polynomial: $\lambda^3 - 14\lambda^2 + 46\lambda + 1 = 0$

For inverse, compute $\det A = -1$ (from constant term)

Adjugate computation gives:

$$A^{-1} = \begin{pmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{pmatrix}$$